

# Current Flow vs Future Flow — Jahna / FastWeigh

**Date:** 2026-05-27 · **For:** Ops + Sales budget meeting (Dividia internal)

**Purpose:** show what the existing scale-house stack does today, what the Dividia + FastWeigh phased model replaces, and what equipment goes in at each phase. Open unknowns called out per phase.

**Owner color key:** Legacy (AxlEye / Mettler Toledo DataBridge MS)

Dividia (new)

FastWeigh (third party)

FDOT (regulator)

Operator / Driver

## 1. Today's flow — what we are replacing

One inbound truck, scale-house perspective. This is the flow Jahna runs today at sites that have AxlEye (4 of 6 — Hilochee + Savannah do not).

**DRIVER** ❶ Truck pulls onto inbound (tare) scale. Driver holds OW permit up to a driver-window camera. Permits are per-permittee (carrier), shown every truck, every time.

**AXLEYE** ❷ Decodes the QR on the permit, calls FDOT's public PAS URL, captures the permit JPG. Saves to scale-PC folder.

**AXLEYE + OPERATOR** ❸ Operator marks first-axle and last-axle lines on the side-view camera feed. AxlEye calculates max allowable gross via bridge formula. Saves XML + JPG to scale-PC folder.

**DATABRIDGE PLUGIN** ❹ DataBridge's AxlEye plugin pulls AxlEye's XML measurement data into the DataBridge transaction. A second plugin renames the XML + JPG with the DataBridge ticket # + timestamp once the ticket is created.

**METTLER TOLEDO DATABRIDGE MS + OPERATOR** ❺ Operator interacts with DataBridge MS on the scale PC: selects hauler / truck / customer / contract, "Get Weight", "Accept". DataBridge reads the Mettler IND780 indicator over serial / Ethernet, creates the ticket, stores everything in its local SQL Server database. Truck marked InYard.

**CUSTOM HTML POLLER** ❻ Customer-written HTML script on each loader tablet polls the scale-PC folder. Reads the renamed XML, displays InYard trucks with their max-weight target to the loader operator.

**METTLER TOLEDO DATABRIDGE MS + OPERATOR** ❼ Truck drives to loading circle, gets loaded. Returns to outbound scale. Operator selects from DataBridge's InYard list, adjusts with clam-shell bucket weight, "Complete". Ticket prints via DataBridge's Crystal Reports module; some customers also emailed.

## Today's stack — what each piece does and where it lives

| Component                                                           | Where                           | What it does                                                | Replaced?                                                                              |
|---------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------|
| AxlEye PC + software                                                | Scale house                     | QR permit reading, axle measurement, bridge formula         | Yes — Dividia replaces (our scope)                                                     |
| DataBridge AxlEye plugin                                            | Scale PC                        | Pulls AxlEye XML into DataBridge ticket                     | Yes — Dividia replaces (our scope)                                                     |
| DataBridge file-rename plugin                                       | Scale PC                        | Renames XML+JPG with ticket# + timestamp                    | Yes — Dividia replaces (our scope)                                                     |
| Mettler Toledo DataBridge MS                                        | Scale PC (Windows + SQL Server) | Ticket creation, printing, Crystal Reports composite        | Yes — <b>Jahna replaces with FastWeigh in 2-3 weeks (their project, not ours)</b>      |
| Scale indicator (Mettler Toledo IND780 assumed; not 100% confirmed) | Wired to scale                  | Scale weight readout to DataBridge                          | No — KEPT (FastWeigh reads it directly after the migration)                            |
| Permit / measurement folder                                         | Scale-PC shared dir             | Drop-zone the loader-tablet HTML polls                      | Yes — replaced by FastWeigh's loader app + our portal                                  |
| Loader-tablet HTML poller                                           | Tablet (customer-written)       | Polls folder, renders InYard trucks + max weight            | Yes — replaced by FastWeigh loader app (per FW go-live)                                |
| Uniview measurement cam (IPC2324SR5-ADZK-G)                         | Side view of scale              | Side-view video for measurement                             | No — REUSED by Dividia                                                                 |
| Uniview QR cam (IPC3635SB-ADZK-I0)                                  | Driver window                   | Driver holds permit up here                                 | No — REUSED by Dividia                                                                 |
| Scales (serial-output indicators)                                   | Inbound + outbound              | Read weight; serial → IP via Moxa NPort 5110 on quote 17938 | No — REUSED. <b>[CORRECTED 2026-06-09]</b> prior "IP-based, no Moxa needed" was wrong. |
| FDOT PAS public lookup                                              | External (pas.fdot.gov)         | Permit verification                                         | No — KEPT (we call directly, public URL)                                               |

## 2. Phase 1 (base) — Dividia + FastWeigh, manual measurement

Same operator workflow, our hardware + software underneath. Manual line placement preserved so the workflow Jahna's operators know does not change on day one.

**DRIVER 1** Truck pulls onto inbound (tare) scale. Driver holds permit up to the Uniview QR camera. Drop axle deployed (mandatory for valid measurement).

**DIVIDIA NVR (RECORDER) 2** Scale-on event fires from IP scale. Recorder opens a TruckTransaction. QR camera frame-decoder reads the permit QR, recognizes `pas.fdot.gov` URL, emits `qr_Read` event.

**FDOT PAS (PUBLIC URL) 3** Recorder's `permit_client` fetches the public permit PDF from `pas.fdot.gov/Public.aspx/ShowPermitDocument/{guid}`. Parses permittee, expiration, axle limits, GVW. No credentials needed.

**JETSON SPOT DECODER + OPERATOR 4** Side-view camera feed renders on the HDMI display next to the scale. Operator marks first / last axle lines using the USB HID controller. Spot Decoder calculates max allowable gross via Florida bridge formula + permit caps. Operator confirms.

**FASTWEIGH + OPERATOR 5** Operator completes their normal FastWeigh ticket entry (hauler / truck / customer / contract / weight / accept). FastWeigh fires `ticket.created` webhook to Dividia cloudapi.

**CLOUDAPI + RECORDER 6** cloudapi receives webhook, HMAC-verifies via Svix, pushes to the on-site recorder via WebSocket. Recorder merges ticket data into the open TruckTransaction (which already has scale + QR + measurement + permit events).

**RECORDER → CLOUDAPI 7** On transaction finish, recorder uploads the whole batch (events + images + permit PDF) to cloudapi. `SW_Transaction` row inserted with summary fields (`bAxleCount`, `fBridgePass`, `bTicket`).

**SCALEWATCHER PORTAL 8** Audit-trail view: every truck's permit PDF, measurement image, alerts (invalid permit, missing QR, overweight), tare ↔ loadout pairing. Cross-site visibility Jahna does not have today.

**FASTWEIGH LOADER APP 9** Loader tablet shows the ticket from FastWeigh. **If Brad confirms FW can surface external measurement**, our max-weight number renders here. Otherwise, see "open unknown" below.

**Open unknown — Phase 1 → loader visibility.** If FastWeigh's loader app cannot surface our max-weight number, options are: (a) push our value into FastWeigh via API so their app renders it, (b) keep a Dividia loader display alongside FastWeigh (adds ~40–60 hr), (c) leave the existing HTML poller running on the tablets pointed at our new output folder (cheapest stopgap; depends on Jahna's willingness to maintain it).

**Open unknown — transaction anchor.** Today, AxIEye runs *before* the DataBridge ticket exists; the AxIEye plugin reconciles them after. Once FastWeigh replaces DataBridge in 2-3 weeks, we need to know whether FastWeigh emits a `ticket.pending` webhook (best), is queryable via GraphQL on demand (good), or neither (we self-anchor on our scale-on event and correlate later by time + truck). All three are workable; cost differs. *Two concurrent migrations happen the same week — Jahna's DataBridge→FastWeigh swap, and our AxIEye+plugins→Dividia swap.*

### 3. Phase 2 (enhanced) — AI auto-detect, same hardware

---

Software-only delta from Phase 1. No new hardware. The Jetson already has GPU headroom from Phase 1 day one — this is the reason we land a Jetson at Phase 1 instead of a cheaper CPU box.

**JETSON SPOT DECODER (AI) A** When truck is stationary on the scale, AI model auto-locates first + last axle, auto-counts axles, computes bridge formula. Operator sees the result with axle markers overlaid.



**OPERATOR B** Reviewer role only. Confirm or override the AI's lines. Same UI as Phase 1, just pre-filled. If override needed, manual line placement still works.



**REST OF THE PIPELINE C** Identical to Phase 1 from this point — permit, ticket, batch upload, portal display.

**Open unknown — Phase 2 training data.** AI needs labeled side-view frames with axle positions marked. Open question: does Jahna's existing AxIEye history give us a starting corpus, can we generate one in early Phase 1 from operator-marked lines, or do we need to label from scratch?

### 4. Phase 3 (speculative) — in-motion read

---

Furthest out, not in the current quote. Listed for context only.

- Front-of-scale camera reads axles as truck rolls onto the scale, before it stops.
- Removes the stationary-frame requirement; operator interaction effectively zero on the happy path.
- Requires camera positioning + lighting work + a velocity-tolerant AI model. Speculative until Phase 2 lands.

## 5. Equipment list per phase

Per-site BOM. Multi-site sites scale linearly except where noted.

| Equipment                                                      | Phase 1 (base)        | Phase 2 (AI)                     | Reliability "Standard"    | Reliability "Failover"             | Reuse from today?                                                                                         |
|----------------------------------------------------------------|-----------------------|----------------------------------|---------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Dividia NVR (on-premise)                                       | 1                     | —                                | —                         | 1 cold spare imaged                | No, new                                                                                                   |
| Jetson Spot Decoder                                            | 1 (Orin Nano / NX)    | — (same unit)                    | —                         | 1 hot spare                        | No, replaces AxIEye PC                                                                                    |
| HDMI display for operator                                      | 1                     | —                                | —                         | —                                  | Possibly existing AxIEye monitor                                                                          |
| Operator input controller — Razer Stream Controller (w/ knobs) | 1 per scale           | — (still useful for AI override) | —                         | 1 spare                            | Yes — REUSE existing at the 4 AxIEye sites; new (\$270 list, ~\$120 used like-new) at Hilochee + Savannah |
| Uniview measurement cam IPC2324SR5-ADZK-G                      | 1 per inbound scale   | —                                | —                         | —                                  | Yes — REUSE                                                                                               |
| Uniview QR cam IPC3635SB-ADZK-I0                               | 1 per inbound scale   | —                                | —                         | —                                  | Yes — REUSE                                                                                               |
| IP scale (existing)                                            | existing              | —                                | —                         | —                                  | Yes — REUSE                                                                                               |
| UPS (per device)                                               | recommended           | —                                | required                  | required + sized for full failover | No, new — site existing UPS to be assessed                                                                |
| Surge protection — Ethernet / serial / power                   | basic                 | —                                | enhanced (per line)       | enhanced + ground audit            | Site-by-site assessment                                                                                   |
| Network drop in scale house                                    | existing if available | —                                | —                         | redundant uplink (LTE or 2nd ISP)  | Possibly existing                                                                                         |
| Pre-imaged USB recovery media                                  | —                     | —                                | 1 NVR + 1 Jetson, on-site | same                               | No, new                                                                                                   |

## 6. Known unknowns — by phase boundary

### Today → Phase 1

- **Loader-tablet display path** — does FastWeigh's loader app surface our measurement, or do we need a Dividia loader display alongside? (FastWeigh call.)
- **Transaction anchor event** — does FastWeigh emit `ticket.pending`, or do we self-anchor on scale-on and correlate later? (FastWeigh call.)
- **Webhook reliability semantics** — replay-on-outage, idempotency, dedup window. Drives Standard vs Failover reliability scope. (FastWeigh call.)
- **Custom UDFs for multi-site routing** — 6 sites, need clean per-site identification on the webhook. (FastWeigh call.)

- **Permit type — divisible vs nondivisible** — affects max-gross cap calculation. Jahna's haulers haul sand (divisible commodity) but sample permit was nondivisible. (Jahna call.)
- **AxlEye sunset timeline** — sets our delivery schedule. (Jahna call.)
- **Day-to-day Jahna technical contact** + Van Fleet site visit window. (Jahna call.)

### Phase 1 → Phase 2

- Training data corpus — derived from Phase 1 operator-marked lines, or labeled from scratch?
- Operator-review UI — confirm/override flow, fallback to full manual on low confidence.
- Acceptance criteria — what detection accuracy % triggers AI-primary vs operator-primary?

### Phase 2 → Phase 3

- Camera positioning for in-motion read — front-of-scale mounting + lighting.
- Truck velocity tolerance of the model.
- Weather + lighting edge cases (rain, dawn / dusk, headlight glare).
- Whether the operator role becomes pure exception-handling or remains supervisory.

---

Sources: cloudapi/docs/jahna-fastweigh/ (architecture.md, open-questions.md, decisions.md, worklog.md), memory model-shift 2026-05-26, jahna-system-replacement-analysis 2026-05-19.